



An anisotropic elastoplastic damage constitutive model for 3D needled C/C-SiC composites



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ABSTRACT

This paper established a combined elastoplastic damage model to analyze the nonlinear mechanical behavior of 3D needled C/C-SiC composites. Inelastic deformation and stiffness degradation of the composite were characterized by the plasticity and damage theories. An innovative plastic potential function containing variable parameters was proposed to consider particularly the anisotropy of plastic deformation in each material direction. Based on the Weibull statistical distribution of the material strength, an exponential damage state function was established to characterize stiffness degradation for the composite in each material direction. Parameters of this constitutive model were determined from experiments data. It can be found that the nonlinear stress-strain curves for the composite under off-axis tensile and shear loadings can be accurately described by the model. The yield and damage surfaces of the composite were also studied. Finally, the constitutive model was validated by analyzing the mechanical behavior of a composite plate containing a center-hole subjected to tensile load.

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1. Introduction

3D needling (or needle-punching) technology can efficiently produce the carbon fiber preforms with low manufacturing costs [1–3]. The current 3D needling technologies for manufacturing advanced composite preforms are reviewed in [4]. The needled carbon fiber preforms can be used to manufacture the C/C and C/C-SiC (or C/SiC) composites with the chemical vapor infiltration (CVI) procedure [5]. Owing to their excellent thermo-mechanical properties at elevated temperatures, the 3D needled composites have been regarded as candidate materials for aerospace components such as aircraft brake, rocket motor nozzle-throats and exit cones etc. The needled composites contain complicated reinforcement fibers and randomly distributed defects, such as voids and micro-cracks, as observed in Refs. [6,7]. These defects in the composite would drive microstructure damages such as matrix cracking, fiber pull-out and breakage under the increasing load. The progressive microstructure damages will cause the nonlinear behavior of the composite before collapse.

Considerable attentions have been paid on the nonlinear mechanical behavior of fibrous C/C and C/C-SiC composites. The effective properties of the composites were also investigated by

the numerical methods in the microscopic scale [8–11]. The microstructures of the needled composites could be taken into account in the microscopic models. But the material properties of constituents, such as matrix, fibers and fiber/matrix interface are difficult to be determined. Thus, it is suitable to study the nonlinear behavior of the composites in the macroscopic scale if the mesoscopic structures of the composites are difficult to be constructed explicitly. The macroscopic mechanical behavior and damage characteristics of 3D needled C/C-SiC composites were studied by extensive experimental methods [12–14]. Some authors added a nonlinear part to the linear part in the macroscopic constitutive model for fiber reinforced composites [15,16]. Many macroscopic models were also proposed based on damage theories to describe the stiffness degradation of fibrous composites [17,18].

The permanent strains and stiffness degradation were both observed in experimental tests of some C/C and C/C-SiC composites. The damage-induced residual strains can be described by using plasticity theory. The combined plasticity-damage theories have been widely adopted to characterize the composite responses [19–23]. Various plastic yield functions have been introduced in the literatures. The forms of plastic potential functions strongly depended on the macroscopic behavior of materials. Sun and Chen [24] proposed the one-parameter plastic potential function for uni-directional laminates with no plastic deformations in longitudinal direction. Plastic potential functions corresponding with the longitudinal, transverse and shear plastic strains were reported

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in [25–28]. The effect of hydrostatic stress on the plastic behavior has also been considered by [29,30]. It should be noted that the parameters in general plastic yield functions are constant, and the isotropic hardening law is always adopted for simplicity. It means that plastic strains in each principle material direction are accumulated proportionally. Furthermore, the subsequent yield surface would expand uniformly in stress space. In fact, the ratio between plastic strains might change during the loading process due to the anisotropic damage in the composite. Ogi [31] proposed a plastic yield function with a variable parameter based on the Sun-Chen model and characterized the nonlinear mechanical behavior of the unidirectional composites very well. The ratio of transverse plastic strain and the shear plastic strain was a variable. However, the plastic strains in fiber direction was neglected in this model. Thus, further study is still required to study the anisotropic plastic strains as well as stiffness degradation of composites with complex fiber reinforcements.

In the present study, the framework of a macroscopic nonlinear constitutive model for 3D needled C/C-SiC composite was introduced. Inelastic deformations and stiffness degeneration of the material were both taken into account in the model. The anisotropy of plastic strain evolution and stiffness degradation in each principle material direction of the composite subjected to complex in-plane loadings were studied. An innovative yield function containing variable parameters was constructed. Based on the Weibull distribution of material strength, an exponential failure probability function was established to characterize the damage state of the composite. Three scalar damage variables were then introduced to characterize the stiffness degradation. Experimental data of in-plane shear and off-axis tensile tests were used to determine the

model parameters. To verify the accuracy of this model, the mechanical response of the composite plate with a center-hole subjected to tensile load were studied by numerical and experimental methods. In the numerical method, the developed macroscopic nonlinear constitutive model was embedded into the material response algorithm by user subroutine Umat in Abaqus software. It was found that the numerical results were good agreement with the experimental data.

2. Experiments and damage characterization of the material

2.1. Material and experiments

The 3D needled C/C-SiC composites were manufactured using needled carbon fiber preforms by the chemical vapor infiltration (CVI), polymer impregnation-pyrolysis (PIP) and liquid silicon infiltration processes (LSI). Firstly, the CVI process was performed at the temperature about 800–1000 °C for 300–700 h to form the porous C/C composites with the density of about 1.5 g/cm³. During the CVI process, the PyC (pyrolytic carbon) interphases with the thickness of about 2 μm were formed to coat the carbon fibers [32]. Then the PIP process was performed to increase the density of the porous C/C composites further. Finally, the LSI process was conducted at about 1650 °C for 1–3 h to form the C/C-SiC composite. In the LSI process, the liquid Si infiltrated in the porous C/C composites and reacted with some amount of carbon matrix to form the SiC matrix [33,34].

In needled preforms, 0° non-woven cloth layers (direction 1), short-cut fiber net layers and 90° non-woven cloth layers (direction 2) were stacked alternatively. To enhance the delamination

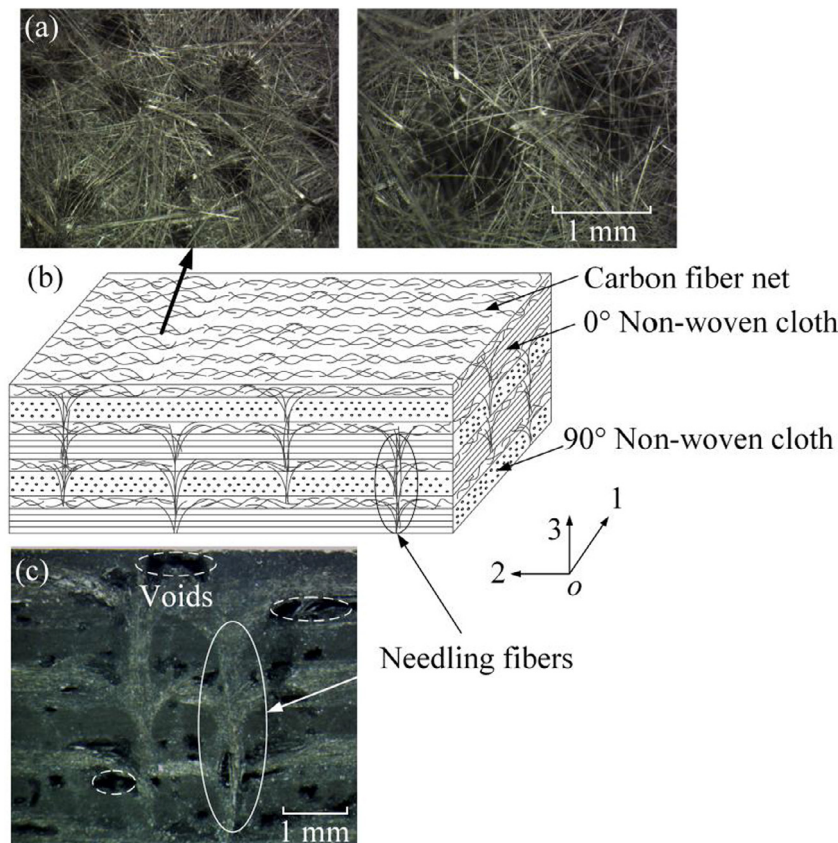


Fig. 1. The microstructure of 3D needled C/C-SiC composite: (a) needling holes on the needled preform, (b) schematic of the needled preform, and (c) needling fibers in the composite.

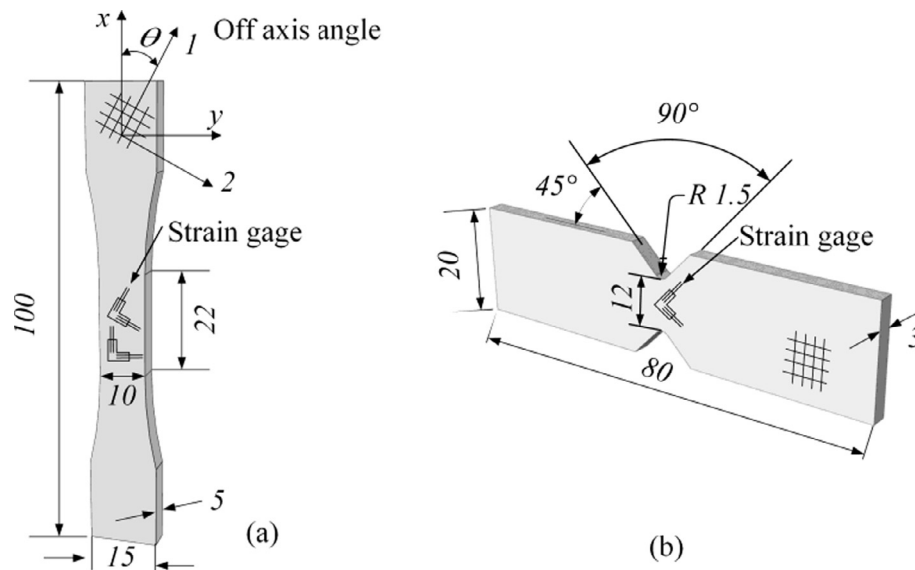


Fig. 2. Test specimens of the composite: (a) tensile specimen and (b) Iosipescu shear specimen.

Table 1
The tested properties of 3D needled C/C-SiC composite.

Specimens	Strength (MPa)	Total failure strain (%)	Plastic failure strain (%)	Poisson's ratio
0° tension	102.63 ± 24.15	0.35 ± 0.12	0.04 ± 0.02	0.12 ± 0.02
15° tension	81.99 ± 19.21	0.38 ± 0.11	0.05 ± 0.01	0.15 ± 0.01
30° tension	69.09 ± 14.53	0.42 ± 0.10	0.07 ± 0.02	0.29 ± 0.02
45° tension	64.84 ± 12.11	0.48 ± 0.12	0.11 ± 0.03	0.39 ± 0.02
Shear	81.19 ± 17.72	1.68 ± 0.14	0.18 ± 0.04	/

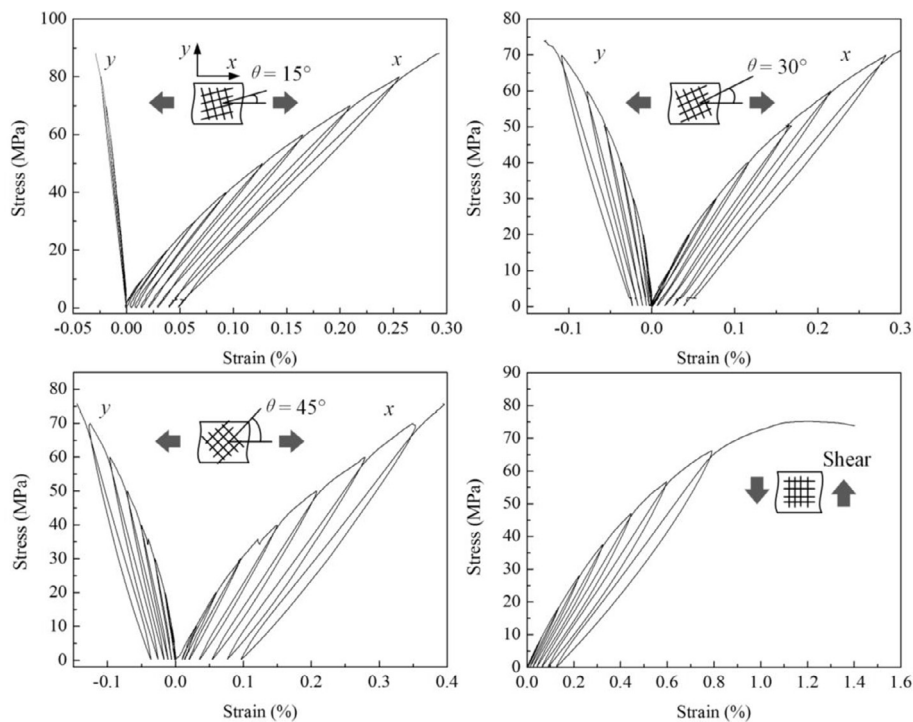


Fig. 3. Stress-strain curves of 3D needled C/C-SiC composite under in-plane tensile and shear loadings in the x - y coordinate system: (a) $\theta = 15^\circ$, (b) $\theta = 30^\circ$, (c) $\theta = 45^\circ$, and (d) shear test.

resistance properties, the preforms were needled by the needles. Some in-plane reinforcement fibers were brought to the thickness direction (direction 3) as shown in Fig. 1. After the needling

process, needling holes were left on the preforms. The fibers in needling holes were deflected and damaged. Moreover, the distribution of needling holes in the preforms were dense and disorder.

Off-axis tensile specimens (with off-axis angle $\theta = 0^\circ, 15^\circ, 30^\circ$ and 45°) and Iosipescu shear specimens were tested under monotonic and cyclic load-unload loadings. Fig. 2 illustrates the schematics of the test specimens. For tensile specimens, the longitudinal and lateral directions are referred to as x and y directions. 0° tensile specimens can also be recognized as on-axis specimens. The specimens with off-axis angle θ are the same with that with off-axis angle $(\pi/2 - \theta)$ due to the symmetry between 1 and 2 directions of the material. Two rectangular strain rosettes were mounted on the center position of the tensile specimen to measure the strains along $x, y, 1$ and 2 directions. One strain rosettes was mounted on the Iosipescu shear specimen at $\pm 45^\circ$ to the loading direction to obtain the engineering shear strain γ_{12} . All the experimental tests were conducted on an Instron-1186 test machine with a crosshead loading speed of 50 MPa/min.

2.2. Mechanical properties

Macroscopic mechanical behavior of 3D needled C/C-SiC composites subjected to in-plane shear and off-axis tensile loadings were investigated. The tested properties of the composite are listed in Table 1, in which the strength and failure strain of tensile specimens were tested along x direction. In off-axis tensile tests, more apparent Poisson's effect can be observed from the specimens with the larger off-axis angle θ . Along the loading direction, the failure strain (both elastic and plastic) increased, while the strength decreased with the increase of θ .

Typical stress-strain curves of off-axis and shear tests for the composite in $x-y$ and 1-2 coordinate systems are shown in Figs. 3–4. All the curves exhibit remarkable nonlinear characteristics. In both tensile and shear cyclic tests, the plastic strain accumulation and elastic modulus degradation can be observed. The stress-strain curves obtained from monotonic loadings approximately envelop the cyclic curves. Thus, we suppose that there is no apparent damage during the unloading-reloading process.

The nonlinear mechanical behavior of C/C and C/C-SiC composites with different reinforcement preforms have been studied in the Ref. [35,36]. Some micromechanical phenomena such as matrix cracking, fiber pull-out and breakage could induce the macroscopic nonlinear behavior of the composite. For the 3D needled C/C-SiC composite, some matrix microcracks could not close completely during the unloading process and the pull-out fibers could not fully return back due to the resistance of friction between the fiber and interface. Therefore, the obvious irretrievable strains could be observed in the cyclic tests.

2.3. Failure mode

The composite failure modes are relevant to the loading conditions. Stress state and failure mode for the tensile and shear specimens are provided in Fig. 5. Under uniaxial tensile load, stress components in principle material directions (1, 2 and shear in 1-2 plane) can be obtained by the stress transformation, which can be written as

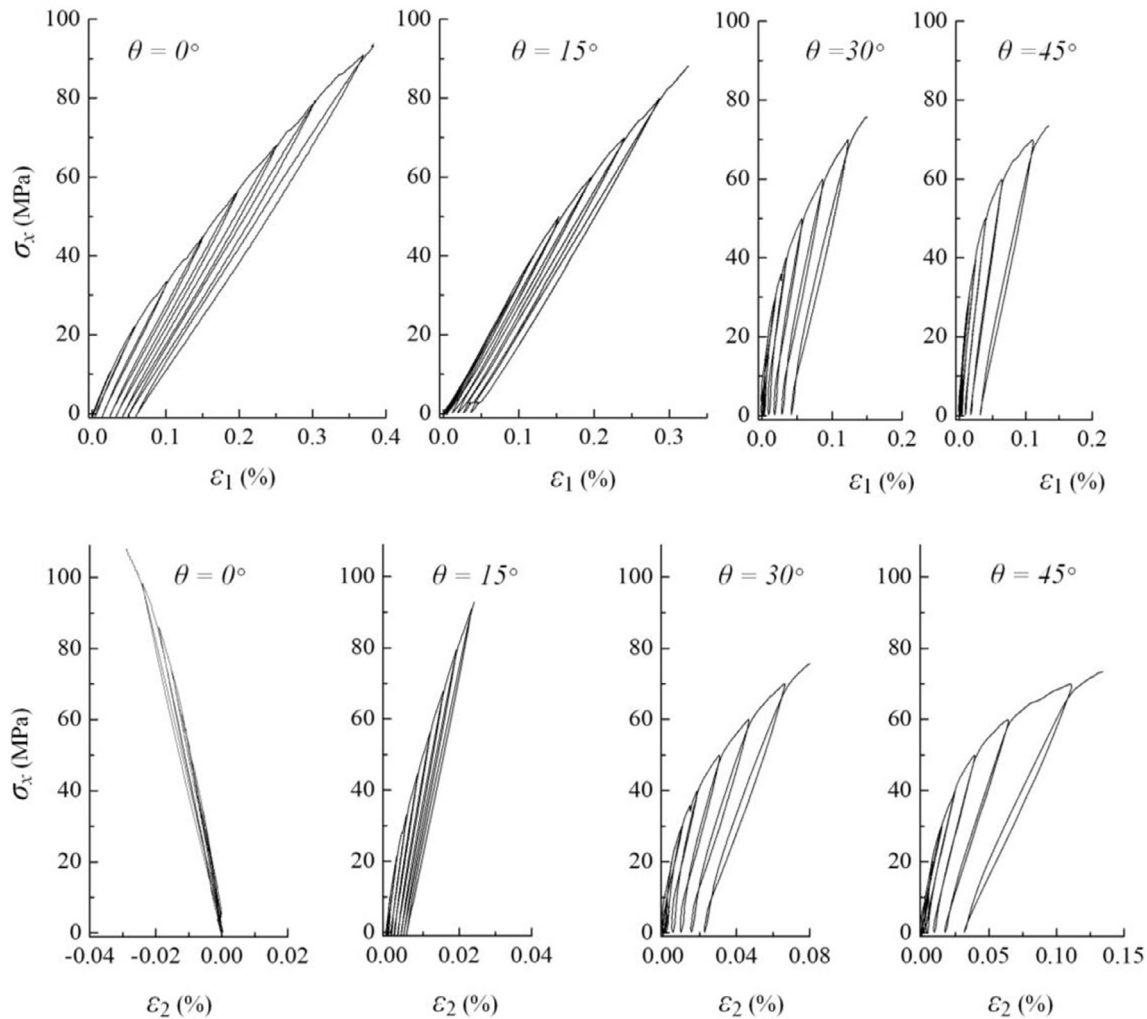


Fig. 4. Strains in principle material directions.

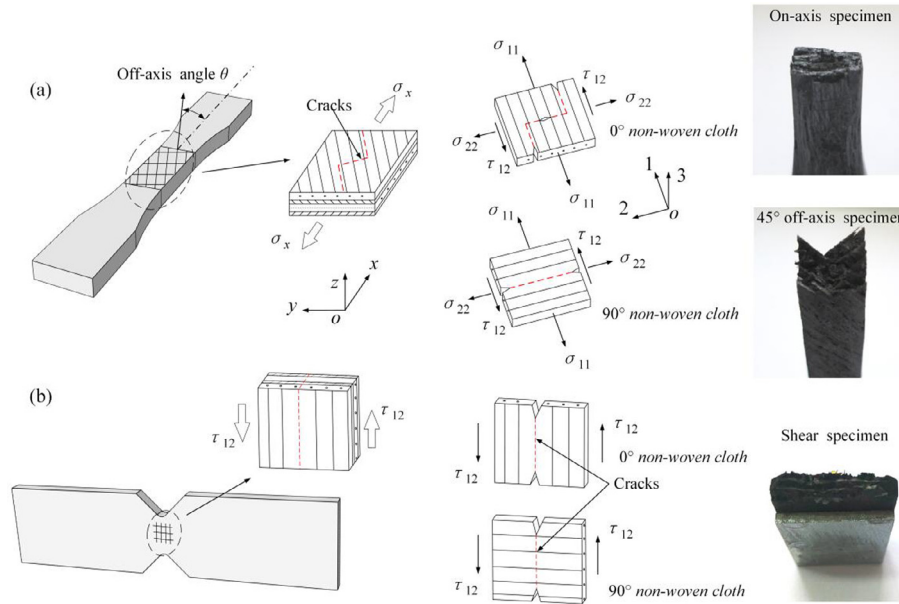


Fig. 5. Stress states and failure modes of the composite specimens: (a) tensile specimen and (b) shear specimen.

$$\begin{cases} \sigma_{11} = \sigma_x \cos^2 \theta \\ \sigma_{22} = \sigma_x \sin^2 \theta \\ \sigma_{12} = -\sigma_x \sin \theta \cos \theta \end{cases} \quad (1)$$

For on-axis tensile specimens, $\sigma_{11} = \sigma_x$ and $\sigma_{22} = \sigma_{12} = 0$, the crack perpendicular to 1 loading direction would initiate and propagate during the tests. As reported by Chen [32], when the cracks approached the carbon fibers, it was difficult to deflect at the interface due to the high interface strength. The carbon fiber would break and only the nearest fibers shared the load during this process. Then the surviving fibers would break one by one due to the high stress concentrations with the increase of loading. Fracture surfaces of on-axis tensile specimens are comparatively smooth. As shown in Fig. 6, fracture surfaces of the non-woven fiber bundles lie in the plane perpendicular to the loading direction and no apparent fibers are pulled out.

Both longitudinal (along fiber direction) and transverse cracking would occur in the non-woven cloth of off-axis specimens as a result of the combination of normal and shear stress. The fracture surface for off-axis specimens are jagged. Delamination and splitting can be observed in the material. Cracks of the non-woven cloth are naturally aligned with the reinforcement fibers. The fracture surfaces of 0° and 90° non-woven cloth are interlaced due to the different arrangement directions of the fibers. Step-shaped failure surfaces of the non-woven fiber bundles can be found in the SEM images. Many cracks can be observed in fiber bundles of the composite under the tensile-shear combined stress.

For the shear specimen, cracks would initiate in cross section at the notch roots. The cracks in the 0° and 90° non-woven cloth propagated along the loading direction. A fracture surface finally connected the two V-shaped notches of the shear specimen. Fiber/matrix debonding are observed in the SEM images of shear specimens. A great amount of single fiber and fiber bundles are pulled out as observed in the fracture surface.

3. Constitutive model

An elastoplastic damage combined constitutive model was established to describe the macroscopic nonlinear behavior of 3D needed composite under various in-plane loading conditions. In

the proposed model, the residual deformations and stiffness degeneration were characterized by the plasticity and damage theories. In the incremental form, the total strain tensor $d\epsilon$ can be decompose into two parts, the plastic part $d\epsilon^p$ and elastic part $d\epsilon^e$, which can be expressed as

$$d\epsilon = d\epsilon^p + d\epsilon^e \quad (2)$$

3.1. Plasticity theory

3.1.1. Plastic yield function with variable parameters

The macroscopic irreversible deformations of the composite can be characterized by the plastic theory. The plastic strain represents all the irreversible deformations induced by the internal damage of the composite. The plastic yield function can be expressed as

$$f(\sigma_{ij}) - \kappa(\bar{\epsilon}_p) = 0 \quad (3)$$

where $f(\sigma_{ij})$ is the plastic potential function, σ_{ij} is the stress component in principle material directions, and $\kappa(\bar{\epsilon}_p)$ is the hardening function written in terms of equivalent plastic strain $\bar{\epsilon}_p$. The plastic strains can be obtained by the associated plastic flow rule

$$d\epsilon_{ij}^p = \frac{\partial f}{\partial \sigma_{ij}} d\lambda \quad (4)$$

in which $d\lambda$ is a scalar proportionality coefficient. In the plane stress case, an innovative plastic potential function is proposed, which can be written as

$$f(\sigma_{ij}) = k \sqrt{a_1 \sigma_{11}^2 + a_2 \sigma_{22}^2 + \sigma_{12}^2} = \bar{\sigma} \quad (5)$$

where a_1 and a_2 are anisotropy parameters characterizing the anisotropic between normal and shear plastic deformations, $\bar{\sigma}$ is the equivalent stress. If the parameters a_1 , a_2 and k are constants, the plastic strains can be rewritten as

$$\begin{cases} d\epsilon_{11}^p = \frac{\partial f}{\partial \sigma_{11}} d\lambda = k^2 \frac{a_1 \sigma_{11}}{\bar{\sigma}} d\lambda \\ d\epsilon_{22}^p = \frac{\partial f}{\partial \sigma_{22}} d\lambda = k^2 \frac{a_2 \sigma_{22}}{\bar{\sigma}} d\lambda \\ d\gamma_{12}^p = \frac{\partial f}{\partial \sigma_{12}} d\lambda = k^2 \frac{\sigma_{12}}{\bar{\sigma}} d\lambda \end{cases} \quad (6)$$

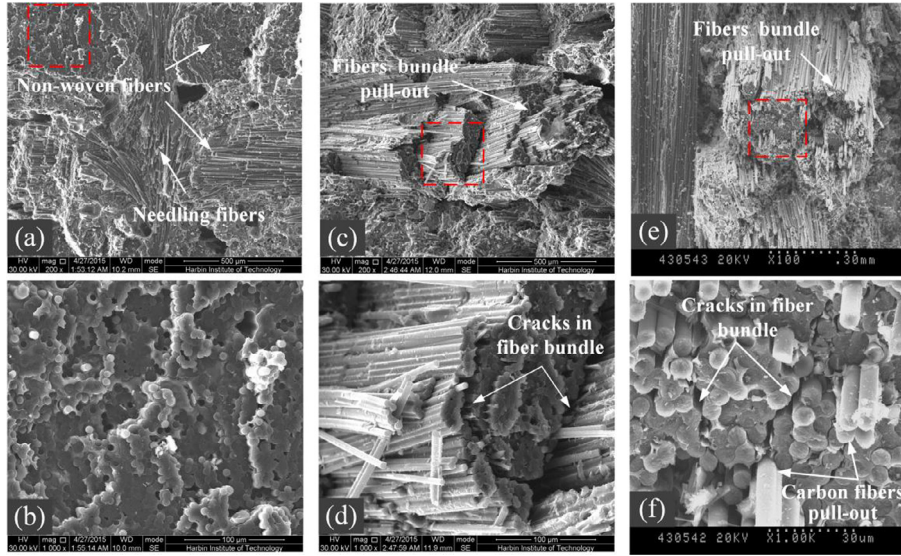


Fig. 6. SEM images of the fracture surfaces: (a), (b) on-axis tensile specimen; (c), (d) 45° off-axis tensile specimen; (e), (f) shear specimen.

$$\begin{cases} \frac{d\epsilon_{11}^p}{d\gamma_{12}^p} = \frac{a_1 \sigma_{11}}{\sigma_{12}} \\ \frac{d\epsilon_{22}^p}{d\gamma_{12}^p} = \frac{a_2 \sigma_{22}}{\sigma_{12}} \end{cases} \quad (7)$$

where, the ratios between normal and shear plastic strains, (i.e. $\frac{d\epsilon_{11}^p}{d\gamma_{12}^p}$ and $\frac{d\epsilon_{22}^p}{d\gamma_{12}^p}$) would remain unchanged during the off-axis loading process. However, the plastic strains in each principle material direction accumulated disproportionately in the off-axis tensile tests due to the anisotropic damage process of the composite. In this work, the values of parameters a_1 and a_2 are related to the loading conditions and would change during the loading process. k is the hardening rate parameter representing the different hardening rate for the composite under various loading conditions. For simplicity, the formulations of plastic strain increment are not changed as given in Eq. (6), but the parameters a_1 , a_2 and k are corrected by using experimental data during the loading process.

3.1.2. Plastic hardening rule

The increment of plastic work per unit volume is written as

$$dW_p = \sigma_{ij} d\epsilon_{ij}^p = \bar{\sigma} d\bar{\epsilon}_p \quad (8)$$

By using Eq. (6), we have

$$\bar{\sigma} d\bar{\epsilon}_p = \sigma_{11} d\epsilon_{11}^p + \sigma_{22} d\epsilon_{22}^p + \sigma_{12} d\gamma_{12}^p = \bar{\sigma} d\lambda \quad (9)$$

Then, the scalar proportionality coefficient $d\lambda$ can be written as

$$d\lambda = d\bar{\epsilon}_p \quad (10)$$

The hardening law of the composite is established by the power law function [24], which can be expressed as

$$\kappa(\bar{\epsilon}_p) = \bar{\sigma} = A(\bar{\epsilon}_p)^n \quad (11)$$

3.1.3. The plastic parameters

By Eq. (7), parameters a_1 and a_2 can be determined using the ratios between normal and shear plastic strain increment, as shown in Eq. (12).

$$\begin{cases} a_1 = \frac{d\epsilon_{11}^p}{d\gamma_{12}^p} \frac{\sigma_{12}}{\sigma_{11}} \\ a_2 = \frac{d\epsilon_{22}^p}{d\gamma_{12}^p} \frac{\sigma_{12}}{\sigma_{22}} \end{cases} \quad (12)$$

In the experimental tests, plastic strain increments $d\epsilon_{11}^p$, $d\epsilon_{22}^p$ and $d\epsilon_x^p$ are measured by the strain gage. The shear plastic strain increment $d\gamma_{12}^p$ can be deduced from the strain transformation formula

$$d\epsilon_x^p = d\epsilon_{11}^p \cos^2 \theta + d\epsilon_{22}^p \sin^2 \theta - d\gamma_{12}^p \cos \theta \sin \theta \quad (13)$$

Under off-axis tensile load, the equivalent stress can be written as the following form by the coordinate transformation of stress components

$$\bar{\sigma} = h(\theta) \sigma_x \quad (14)$$

$$h(\theta) = k \sqrt{a_1 \cos^4 \theta + a_2 \sin^4 \theta + \cos^2 \theta \sin^2 \theta} \quad (15)$$

By Eqs. (6), (10) and (15) we have

$$d\epsilon_x^p = h(\theta) d\bar{\epsilon}_p \quad (16)$$

In the case of proportional loading, the equivalent plastic strain can be expressed as

$$\bar{\epsilon}_p = \frac{\epsilon_x^p}{h(\theta)} \quad (17)$$

In the in-plane pure shear case, the hardening rate parameter was set as $k = 1$. The equivalent stress and equivalent plastic strain can be written as

$$\bar{\sigma} = k_s \sigma_{12} = \sigma_{12} \quad (18)$$

$$\bar{\epsilon}_p = \frac{\gamma_{12}^p}{k_s} = \gamma_{12}^p \quad (19)$$

The $\bar{\sigma} - \bar{\epsilon}_p$ relationship of the composite can be determined by the $\sigma_{12} - \gamma_{12}^p$ curves obtained from the cyclic shear tests. $\bar{\sigma} - \bar{\epsilon}_p$ relationship of the composite is considered to be identical and independent of loading conditions. Therefore, the values of hardening rate parameter k for different loading conditions can be determined by a master curve, which can be fitted by all $\bar{\sigma} - \bar{\epsilon}_p$ curves of shear and off-axis tensile tests, as illustrated in Fig. 7. So the hardening rate parameter k for 15°, 30° and 45° tensile tests take the value of 1.10, 1.21 and 1.34 respectively. By using the $\bar{\sigma} - \bar{\epsilon}_p$ relationship, the parameters A and n in Eq. (11) also can be obtained. Then the hardening law function of the composite was determined. The evolution of a_1 and a_2 for the composite obtained

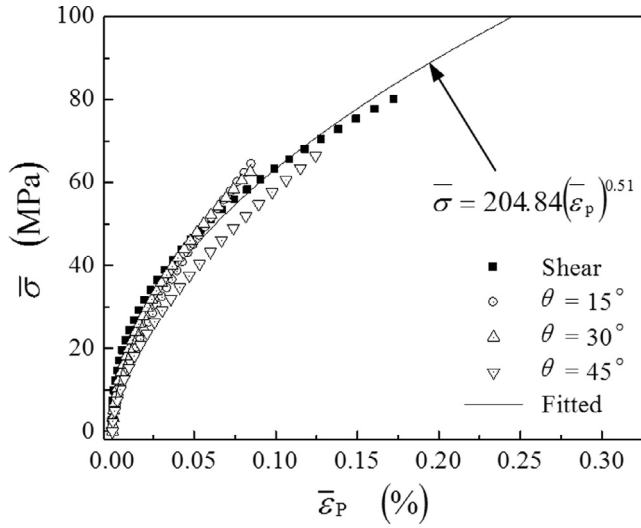


Fig. 7. The $\bar{\sigma} - \bar{\epsilon}_p$ relationship of the composite.

from different off-axis tensile tests are provided by the scatter points in Fig. 8. Due to the symmetry between 1 and 2 directions of the composite, the 15° and 30° off-axis tensile tests can also be referred to as 75° and 60° tests. Therefore, parameter a_1 of 60° and 75° off-axis tests have the same values as a_2 of 30° and 15° off-axis tests, respectively.

As displayed in Fig. 8, the anisotropic parameters a_1 and a_2 closely relate to the loading conditions and change continuously with the load. Thus, a_1 and a_2 can be expressed by the following expressions

$$\begin{cases} a_1 = \psi(r_1, \bar{\sigma}) \\ a_2 = \psi(r_2, \bar{\sigma}) \end{cases} \quad (20)$$

where $\bar{\sigma}$ is the equivalent stress, r_1 and r_2 are loading condition parameters, which can be written as

$$\begin{cases} r_1 = \frac{\sigma_{11}}{\sqrt{\sigma_{11}^2 + \sigma_{22}^2}} + \frac{\sigma_{22}}{\sqrt{\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{12}^2}} \\ r_2 = \frac{\sigma_{22}}{\sqrt{\sigma_{11}^2 + \sigma_{22}^2}} + \frac{\sigma_{11}}{\sqrt{\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{12}^2}} \end{cases} \quad (21)$$

where stress components σ_{11} , σ_{22} and σ_{12} take the positive values. Because of the symmetry between 1 and 2 directions of the composites, the anisotropic parameters a_1 and a_2 can be formulated by the same function expression ψ , which is defined as a power law function

$$\begin{cases} a_1 = \psi(r_1, \bar{\sigma}) = A(r_1)(\bar{\sigma})^{B(r_1)} + C(r_1) \\ a_2 = \psi(r_2, \bar{\sigma}) = A(r_2)(\bar{\sigma})^{B(r_2)} + C(r_2) \end{cases} \quad (22)$$

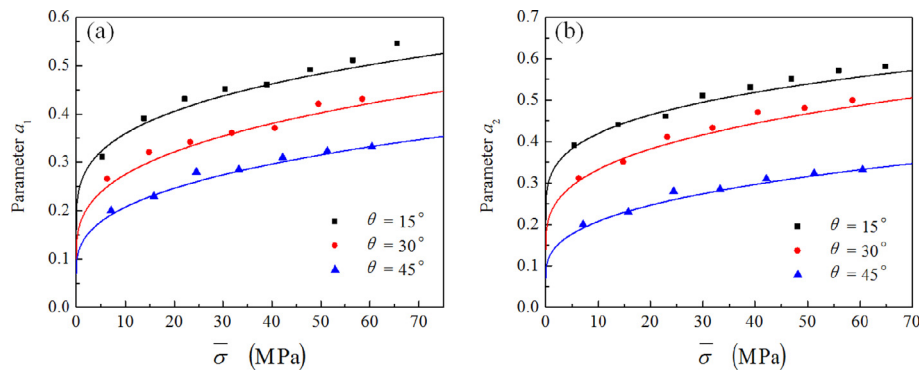


Fig. 8. (a) $a_1 - \bar{\sigma}$ curves and (b) $a_2 - \bar{\sigma}$ curves.

where $A(r_i)$, $B(r_i)$ and $C(r_i)$ ($i = 1, 2$) can be determined by fitting the experimental $a_i - \bar{\sigma}$ relationships, as plotted in Fig. 8. Furthermore, $A(r_i)$, $B(r_i)$ and $C(r_i)$ can be expressed as the linear functions of r_i (Eq. (23)), as shown in Fig. 9(a).

$$\begin{cases} A(r_i) = A_0 + A_r r_i \\ B(r_i) = B_0 + B_r r_i \\ C(r_i) = C_0 + C_r r_i \end{cases} \quad i = 1, 2 \quad (23)$$

The value of the parameter k has a significant influence on the hardening rate of the composite. The larger value of k represents the faster hardening rate of the composite under the given hardening rule (Eq. (11)). As illustrated by the experimental results, the hardening rate parameter k is related to the complexity of loading conditions. This paper defined a tensile-shear combination parameter r_{TS}

$$r_{TS} = \frac{\sigma_{11} + \sigma_{22} + \sigma_{12}}{\sqrt{\sigma_{11}^2 + \sigma_{22}^2 + \sigma_{12}^2}} \quad (24)$$

Then, a linear functional relationship (Eq. (25)) can be found between k and r_{TS} , as shown in Fig. 9(b).

$$k(r_{TS}) = k_0 + k_r r_{TS} \quad (25)$$

All parameters in the model can be determined by the experiments, which are summarized in Table 2.

3.2. Damage theory

3.2.1. Damage variables

As observed from the fracture surfaces of the test specimens, the cracks in the material were naturally aligned with the fibers. The fracture surfaces of reinforcement fiber bundles lied in planes perpendicular or parallel to fiber directions. So, the orthotropic damage model is adopted in the present work. Secant modulus degradation can be observed from cyclic tests. The damage variables in principle material directions can be written as

$$\begin{cases} d_1 = 1 - E_1/E_1^0 \\ d_2 = 1 - E_2/E_2^0 \\ d_6 = 1 - G_{12}/G_{12}^0 \end{cases} \quad (26)$$

where d_1 , d_2 and d_6 denote the damage developed in 1, 2 and in-plane shear directions. E_1^0 , E_2^0 and G_{12}^0 are initial modulus of the composite, E_1 , E_2 and G_{12} are modulus of the damaged composite.

The plastic strain accumulation and stiffness degeneration can be attributed to the micro-damage of the composite, such as matrix cracking, fiber pull-out and breakage. The damage variables are proportional to the plastic strains for a needled C/C composite under tensile and shear load as studied in [37]. The evolution of damage variables and plastic strains have the same tendency. Thus,

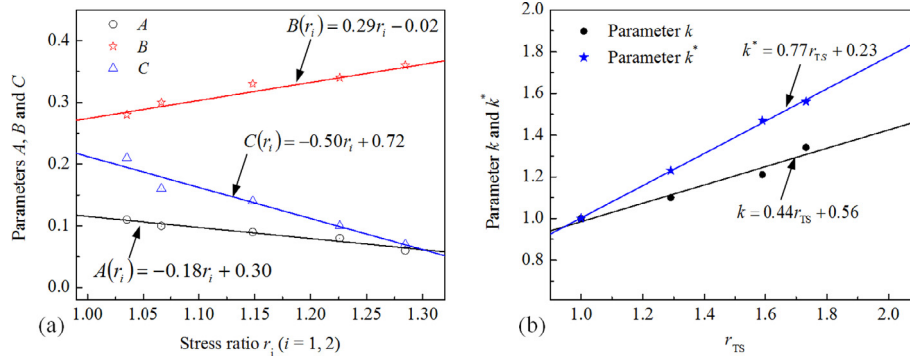


Fig. 9. (a) The relationship between parameters A, B, C and r_i ; (b) the relationship between k , k^* and r_{TS} .

Table 2

Parameters of the proposed model.

Parameters	A (MPa)- n	n	A_r	A_0	B_r	B_0	C_r
Values	204.84	0.51	-0.18	0.30	0.29	-0.02	-0.05
Parameters	C_0	σ_e (MPa)	n_e	k_r	k_0	k_r^*	k_0^*
Values	0.72	90.48	1.67	0.44	0.56	0.77	0.23

the evolution law of damage variables can be expressed by Eq. (27) with the similar form of Eq. (6).

$$\begin{cases} d_1 = k^{*2} \frac{a_1 \sigma_{11}}{\bar{\sigma}^*} D(\bar{\sigma}^*) \\ d_2 = k^{*2} \frac{a_2 \sigma_{22}}{\bar{\sigma}^*} D(\bar{\sigma}^*) \\ d_6 = k^{*2} \frac{\sigma_{12}}{\bar{\sigma}^*} D(\bar{\sigma}^*) \end{cases} \quad (27)$$

$\bar{\sigma}^*$ is the damage equivalent stress and can be written as

$$\bar{\sigma}^* = k^* \sqrt{a_1 \sigma_{11}^2 + a_2 \sigma_{22}^2 + \sigma_{12}^2} \quad (28)$$

$D(\bar{\sigma}^*)$ represents the damage state of the composite which increase monotonously with the loadings. a_1 and a_2 are anisotropic parameters which take the same value as in the plastic potential function. It is assumed that the shape of damage and yield surfaces for the composite were the same during the loading process. k^* relates to the damage rate of the composite and is referred to as the damage rate parameter.

3.2.2. The damage parameters

Under in-plane pure shear load, we set $k^* = 1$, and the $\bar{\sigma}^* - D(\bar{\sigma}^*)$ curve for the composite can be determined by the $\sigma_{12} - d_6$ relationships from the shear tests.

By Eq. (27), we have

$$D(\bar{\sigma}^*) = \frac{d_1 \sqrt{a_1 \sigma_{11}^2 + a_2 \sigma_{22}^2 + \sigma_{12}^2}}{k^* a_1 \sigma_{11}} \quad (29)$$

The damage state $D(\bar{\sigma}^*)$ can be derived by d_1 and k^* from the off-axis tensile tests. It is assumed that the relationship between $\bar{\sigma}^*$ and $D(\bar{\sigma}^*)$ for each off-axis test is coincident with that obtained from shear tests. So, the values of k^* for different tests can be determined by numerical fitting. The $\bar{\sigma}^* - D(\bar{\sigma}^*)$ curves for the composite are displayed in Fig. 10, in which the values of k^* are 1.00, 1.23, 1.47 and 1.56 corresponding with shear, 15° tension, 30° tension and 45° tension, respectively. In this paper, k^* can be expressed as the linear function of tensile-shear combination parameter r_{TS} (Eq. (30)), as shown in Fig. 9(b).

$$k^*(r_{TS}) = k_0^* + k_r^* r_{TS} \quad (30)$$

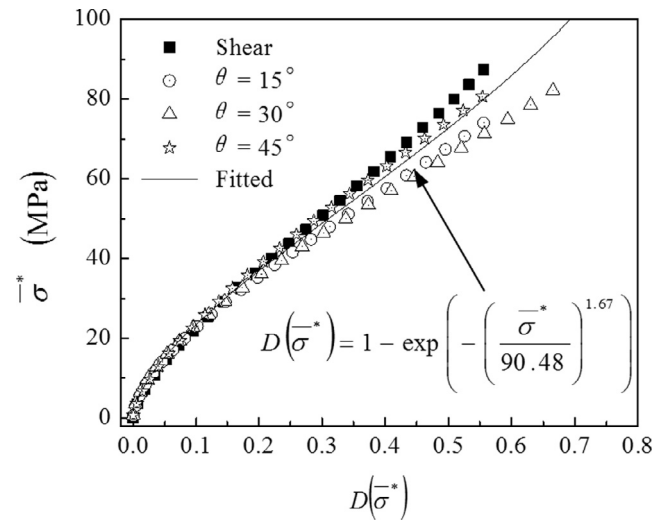


Fig. 10. The $\bar{\sigma}^* - D(\bar{\sigma}^*)$ relationship of the composite.

For the 3D needled composites, the experimental data exhibits some scatter due to the in-homogeneous characteristics of the material. As studied in our previous work [38], the dispersion of the composite performances can be attributed to the random distribution of defects such as fiber breakage at needling positions, matrix microcracks and voids. The scatter of ultimate strength for the composite can be characterized by a Weibull statistical distribution function [39,40], expressed as

$$F(\bar{\sigma}^*) = 1 - \exp\left[-(\bar{\sigma}^*/\sigma_0)^m\right] \quad (31)$$

where $\bar{\sigma}^*$ is the damage equivalent stress, $F(\bar{\sigma}^*)$ is the failure probability of the composite under load $\bar{\sigma}^*$, σ_0 is the scale parameter, m is the Weibull modulus. The larger value of parameter m represents the less scatter of the tested strength. The relationship between $\bar{\sigma}^*$ and $F(\bar{\sigma}^*)$ for the composite is provided in Fig. 11. The failure probability $F(\bar{\sigma}^*)$ could be recognized as the measurement of the damage state of the composite under load $\bar{\sigma}^*$. Therefore, $D(\bar{\sigma}^*)$ can be written in the same form

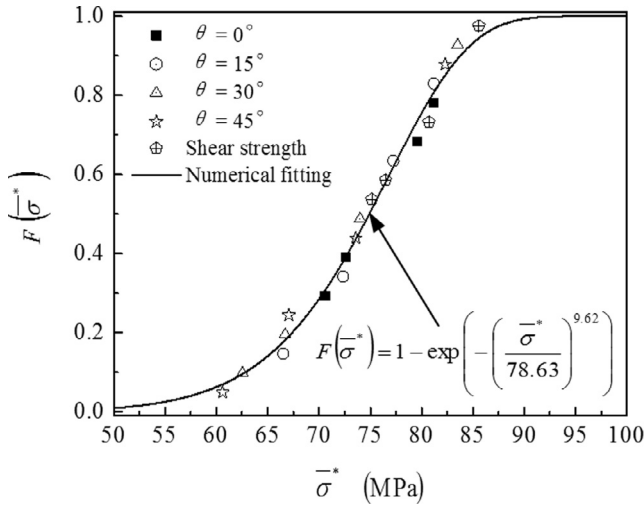


Fig. 11. Weibull distribution of the composite strength.

$$D(\bar{\sigma}^*) = 1 - \exp[-(\bar{\sigma}^*/\sigma_e)^{n_e}] \quad (32)$$

The parameters σ_e and n_e can be obtained by numerical fitting of the experimental data in Fig. 10. All the parameters for the evolution of damage variables are listed in Table 2.

3.2.3. Stiffness matrix for the damaged composite

If the damage variables are determined, the relationship between elastic strain and stress can be presented in the following form

$$\boldsymbol{\varepsilon}^e = \mathbf{C}_d : \boldsymbol{\sigma} \quad (33)$$

where the bold-face symbols $\boldsymbol{\varepsilon}^e$ and $\boldsymbol{\sigma}$ represent elastic strain and Cauchy stress tensor, respectively, $\mathbf{C}_d$ represents the compliance matrix of the damaged composite. The explicitly form of $\mathbf{C}_d$ for orthotropic material is expressed as Eq. (34) [25]. It is assumed that

the components $-v_{12}^0/E_1^0$ and $-v_{21}^0/E_2^0$ in compliance matrix keep constant [37].

$$\mathbf{C}_d = \begin{bmatrix} \frac{1}{E_1^0(1-d_1)} & -\frac{v_{21}^0}{E_2^0} & 0 \\ -\frac{v_{12}^0}{E_1^0} & \frac{1}{E_2^0(1-d_2)} & 0 \\ 0 & 0 & \frac{1}{G_{12}^0(1-d_6)} \end{bmatrix} \quad (34)$$

An equivalent form of the elastic strain and stress relationship can be written as

$$\boldsymbol{\sigma} = \mathbf{S}_d : \boldsymbol{\varepsilon} \quad (35)$$

$$\mathbf{S}_d = \frac{1}{t} \begin{bmatrix} E_1^0(1-d_1) & v_{21}^0 E_1^0(1-d_1)(1-d_2) & 0 \\ v_{12}^0 E_2^0(1-d_1)(1-d_2) & E_2^0(1-d_2) & 0 \\ 0 & 0 & t G_{12}^0(1-d_6) \end{bmatrix} \quad (36)$$

where $\mathbf{S}_d$ represents the damaged stiffness matrix of the composite and the parameter t is expressed as

$$t = 1 - v_{12}^0 v_{21}^0 (1-d_1)(1-d_2) \quad (37)$$

3.3. Results and discussions

3.3.1. The plastic yield surface and damage surface

The parameters a_1 , a_2 , k and k^* can be determined by the loading condition and tensile-shear combination parameters. We can plot the yield surface and damage surface of the proposed model in the $(\sigma_{11}, \sigma_{22}, \sigma_{12})$ stress space by the expressions of $\bar{\sigma}$ and $\bar{\sigma}^*$. The macroscopic performance of the composite exhibits some scattered. Accordingly, the damage equivalent stress $\bar{\sigma}^*$ at the failure point of the composite distributed from 60 MPa to 85 MPa as shown in Fig. 11. This paper plots the yield and damage surface of the proposed model under the load state of $\bar{\sigma} = 50$ MPa and $\bar{\sigma}^* = 50$ MPa as seen in Fig. 12(a) and (b), respectively. For the in-plane tensile-shear case, the surface plots are illustrated in the first

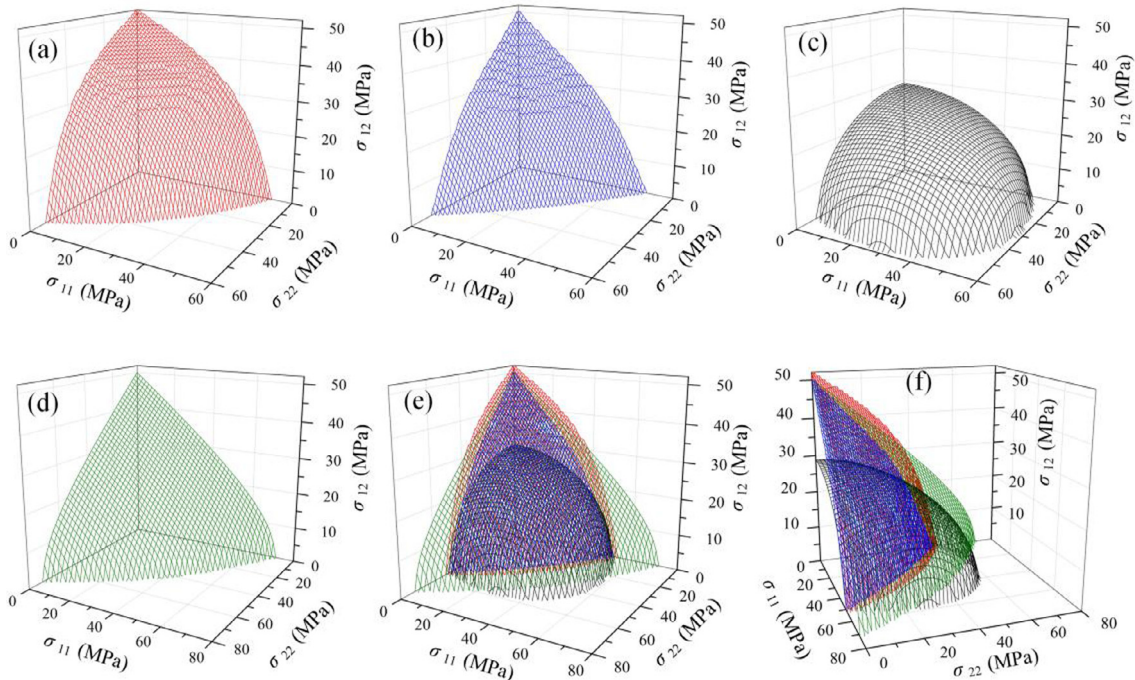


Fig. 12. Plastic yield surface and damage surface of the proposed model compared with other models. (a) and (b): plastic yield surface and damage surface of the proposed model; (c) Mises yield surface; (d) yield surface of Ref. [33]; (e) and (f) comparison of the four surface plots by front view and side view.

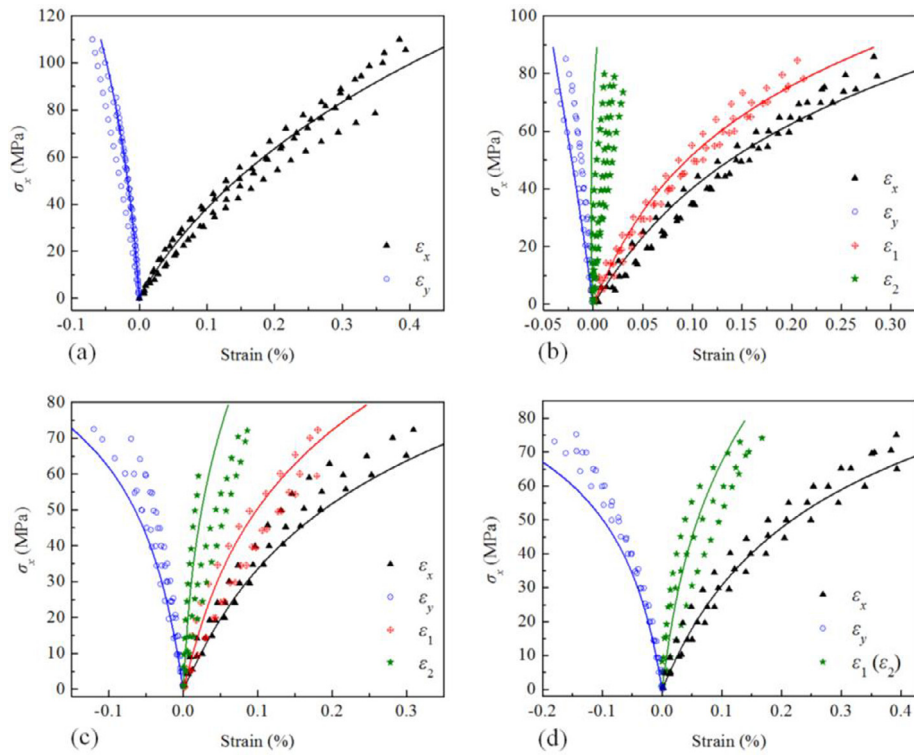


Fig. 13. The predicted off-axis tensile stress-strain curves compared with experimental results: (a) $\theta = 0^\circ$, (b) $\theta = 15^\circ$, (c) $\theta = 30^\circ$ and (d) $\theta = 45^\circ$.

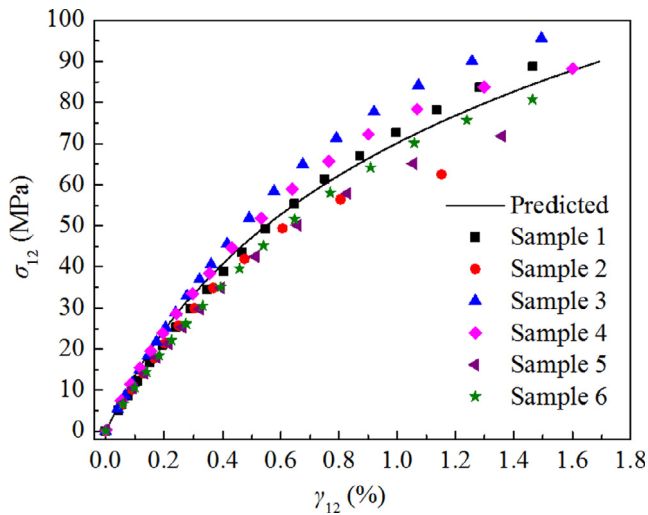


Fig. 14. The predicted shear stress-strain curve compared with experimental results.

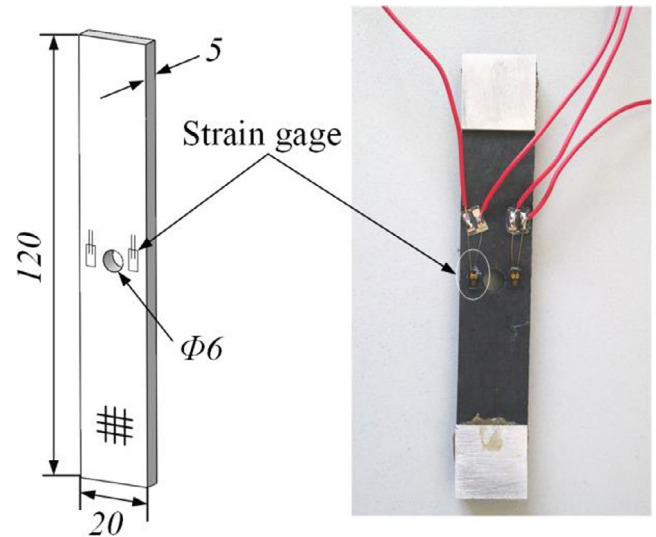


Fig. 15. Composite plate with a through hole in the center.

Euclidean quadrant. Complete yield or damage surfaces would be obtained with the compression-shear behavior of the composite being studied in our further work. For the proposed model, anisotropy parameters (a_1 and a_2) in $\bar{\sigma}$ and $\bar{\sigma}^*$ take the same value, while loading condition parameter k takes a lower value than k^* . Thus, plastic yield surface of the model is slightly larger than the damage surface and the shape of the two surface plots are almost identical.

Mises plastic yield surface is also added in this paper for comparison. Fig. 12(e) provided the Mises yield surface under the stress state of $\sigma_M = 50$ MPa in the $(\sigma_{11}, \sigma_{22}, \sigma_{12})$ stress space, where σ_M denotes the Mises equivalent stress. The shape of Mises yield

surface is distinctively different from that of the proposed model. As displayed in Fig. 12, yield surfaces of Mises yield function and this model cross each other. Mises yield function does not contain the hydrostatic stresses, thus it overrates the effect of shear stress on the yielding of the composite. For pure (or approximate pure) shear case of the composite, Mises function is much more conservative than the proposed model. For tensile-shear combined loading case, Mises yield function seems to be risky.

In our previous work [38], we formulated a plastic yield function (written as Eq. (38)) to describe the axial inelastic behavior of 3D needled C/C-SiC composite. The yield surface of $\sigma_H = 50$ MPa is provided in Fig. 12(d).

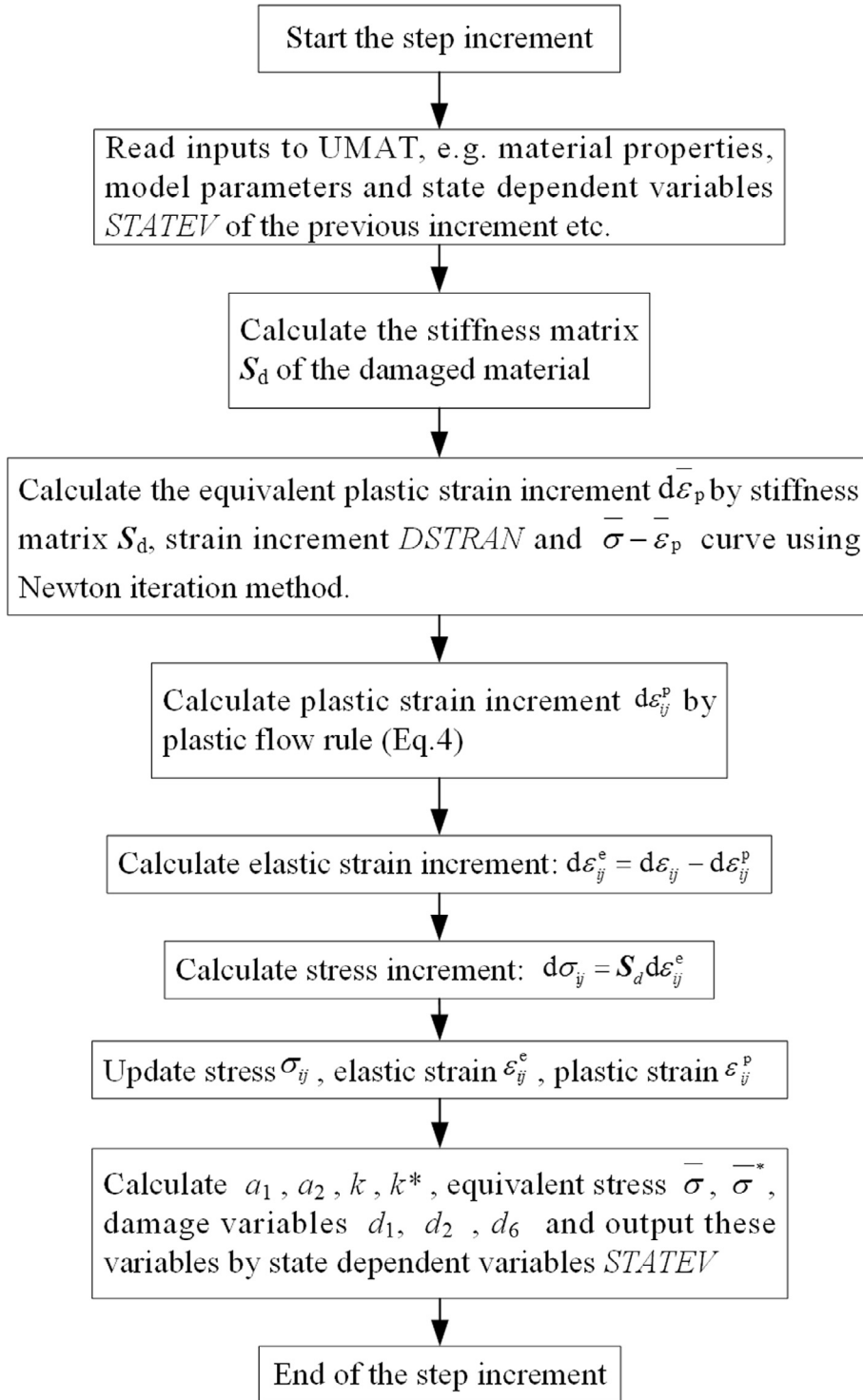


Fig. 16. The flow chart of FEM increment.

$$\sigma_H = \sqrt{0.04(\sigma_{11}^2 - \sigma_{11}\sigma_{22} + \sigma_{22}^2) + \sigma_{12}^2} + 0.50(\sigma_{11} + \sigma_{22}) \quad (38)$$

Eq. (38) was based on the Hill-type yield criterion and considered the influence of hydrostatic stresses on material plasticity. Closing to the σ_{12} axis, the yield surface of σ_H is close to yield and damage surfaces of the proposed model, while far away from σ_{12} axis, yield surface of σ_H is much larger than the present two surfaces. It can be observed that yield function σ_H underestimated the effect of tensile-shear combination on the yielding of the

composite. In the present work, the combination of tensile and shear stress would accelerate the yielding or damage for the composite.

3.3.2. Stress-strain curves

After the model parameters are determined, the mechanical response of the composite subjected to various loading conditions can be obtained by the explicit formulations of elastic and plastic strains. The prediction of stress-strain relationships of the

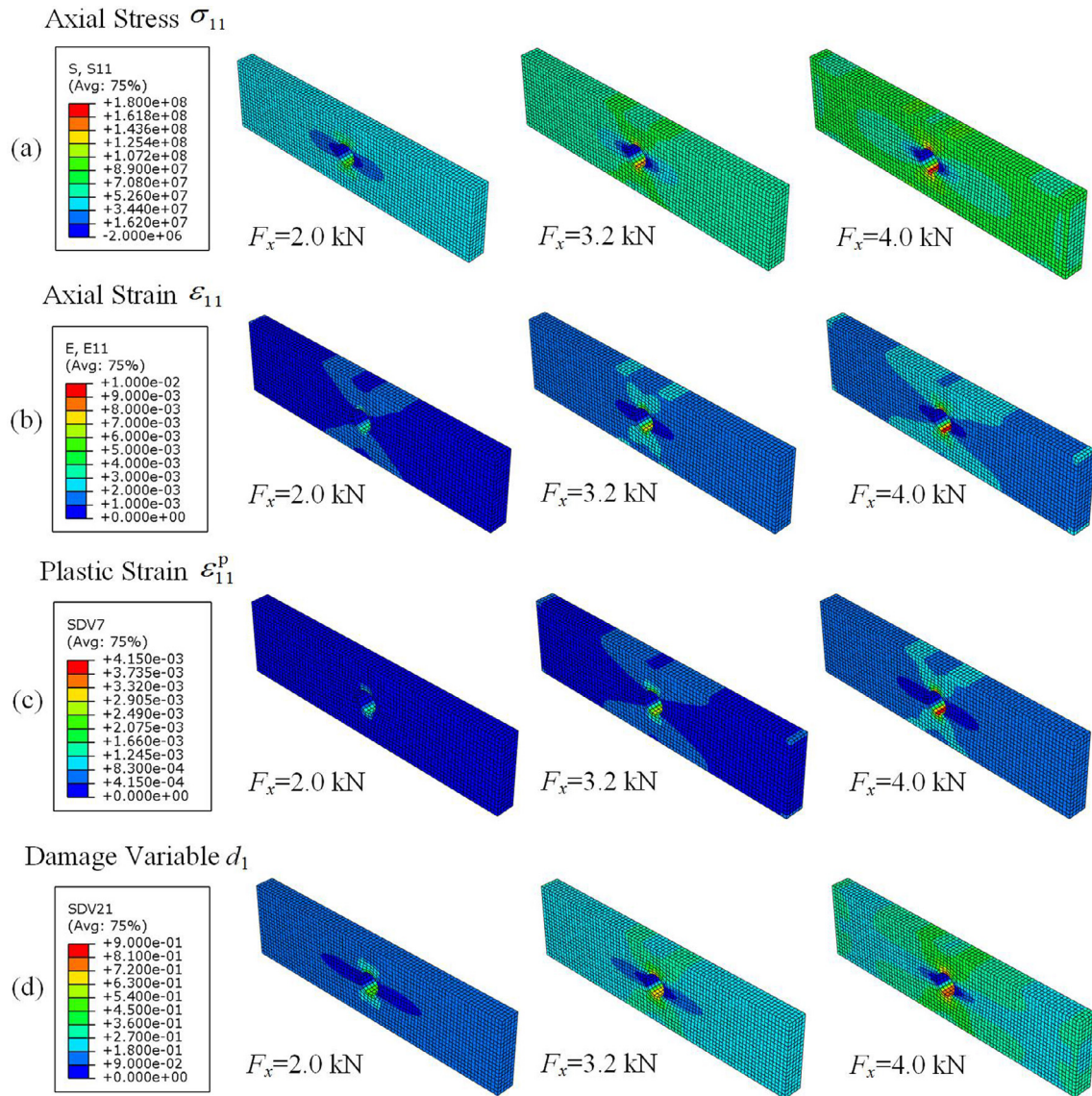


Fig. 17. The contours of FEM calculated results for the composite plate under uniaxial tensile load: (a) σ_{11} , (b) ε_{11} , (c) ε_{11}^p , (d) d_1 .

composite under in-plane tensile or shear load are carried out by the Matlab code. Figs. 13 and 14 display the comparison between the predicted results and the experimental data of off-axis tensile and shear tests. Stress-strain curves of the composite in both x - y and 1-2 coordinate systems are provided. Scatter points in the figures represent experimental data and the predicted results are drawn by solid lines. Both tensile and shear experiment data of the composite all show some scatter. It can be found that the predicted curves are good agreement with the experimental results. The model slightly underestimated the strain ε_2 of the 15° off-axis specimens. In fact, the strain ε_2 for 15° off-axis specimen is a small value, which is difficult to be measured during the experimental measurement. Some errors also may be introduced to measure the strain ε_2 for 15° off-axis specimen.

3.3.3. Experimental validation of the model

The tensile experiment of the composite plate with a through hole in the center was conducted to validate the constitutive model proposed in the present study. Tensile experiment for open-hole specimen of the composite plate was performed on the Instron-1186 test machine. Two strain gages were mounted on the

specimen beside the hole to measure the strains during the test, as shown in Fig. 15. To avoid the sliding between the clamp and specimen, we pasted aluminum sheets (20 mm × 20 mm) on the two ends of the specimen. The tensile load was applied along 0° fiber direction. Both monotonic and cyclic load-unload tests were performed. The load (or unload) speed was 0.3 mm/min. The mechanical behavior for the open-hole specimen of the composites are also simulated by using the present constitutive model, which has been embedded into the user defined material subroutine (Umat) of Abaqus. The numerical model of the composite plate have 12080 C3D8 elements. The two clamped ends of the specimen are not included in the FE model. One end of the model is fixed and the other end is applies with the load of $F_x = 4$ kN incrementally, corresponding to the average critical failure load of the specimens. During the finite element method (FEM) analysis, variables such as $\bar{\sigma}$, $\bar{\sigma}^*$, ε_{ij}^e , ε_{ij}^p and d_i of the model are updated by the end of each increment. Flow chart of the FEM increment is provided in Fig. 16.

Mechanical behavior of the open-hole specimen under incrementally tensile load is simulated by FEM using the proposed constitutive model. In Fig. 16, FEM calculation results including the distribution of σ_{11} , ε_{11} , ε_{11}^p and d_1 of the model under tensile load

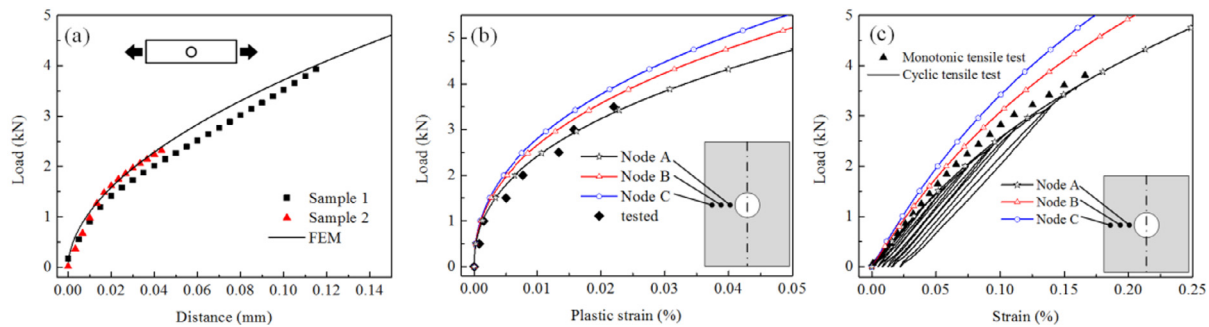


Fig. 18. The numerical calculation results for the composite plate compared with experimental data: (a) load-displacement curves, (b) plastic strain ε_{11}^p and (c) strain ε_{11} .

of 2 kN, 3.2 kN and 4 kN (50%, 80% and 100% of the specimen failure load) are provided. There exists obvious stress concentration around open-hole of the specimen. The specimen collapsed at the center-hole position as observed in experimental tests. The plastic deformation and progressive damage contour of the specimen with the increase of the load F_x are displayed in Fig. 17. It can be found that the distributions of ε_{11} , ε_{11}^p and d_1 of the specimen are also significantly affected by the center-hole. The apparent elastic and plastic deformation, as well as material damage can be observed at the center-hole position of the specimen.

As shown in Fig. 18(a), the calculated load-displacement relationship of the specimen exhibits acceptable accuracy compared with experimental results. Due to the stress concentration effect, strain distribution around the center hole is extremely non-uniform. Three nodes near the center hole on the FE model namely node-A, B and C with the distance of 1, 3, and 5 mm from the hole edge, respectively, are utilized to analyze. The evolution of strain ε_{11} and plastic strain ε_{11}^p versus load for the three nodes are provided in Fig. 18(b) and (c). The location of node-A corresponding to the strain gage position on the specimen. It can be found that the predicted results of node-A are in good agreement with the experimental results. Thus, the proposed constitutive model can characterize the macroscopic mechanical response of the composites subjected to the tensile and shear loading.

4. Conclusions

Based on the damage characteristics and macroscopic mechanical behavior of the material under shear, on- and off axis tensile load, an anisotropic constitutive model for 3D needled C/C-SiC composite was proposed. Inelastic strains of the composite were described by the plasticity theory in a macroscopic perspective. An innovative plastic yield function containing variable parameters was proposed to describe the anisotropic plastic strains. Three scalar damage variables were introduced to characterize the degradation of material modulus. Plastic yield surface and damage surface for the composite were constructed. The proposed model can predict the nonlinear stress-strain behavior of the composite under complex loading conditions and analyze quantitatively the anisotropy of composite damage. The proposed constitutive model embedded into Abaqus Umat is utilized to simulate the open-hole specimen of the composites under uniaxial tensile loading. The predicted results are in good agreement with the experimental results.

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